Lab Manual for Digital Logic and Design

LAB-10

Implementation of 2-to-4 decoder and 4-to-2 encode

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ority encoder

Experiment no.: 07

(a) **Implementation of 2-to-4 decoder and 4-to-2 encoder**

(b) **Implementation of 4-input pri**

To understand and implement a 2-to-4-line decoder and 4-to-2-line encoder. To understand theapplications of decoders and encoders.

Apparatus:

- Logic trainer/ Multisim Software
- Connecting wires
- IC: 7404, 7408, 7432
- Power supply

Theory:

Decoder:

The process of taking some type of code and determining what it represents in terms of a recognizable number or character is called decoding. A decoder is a combinational logic circuit that performs the decoding function, and produce an output that indicates the (meaning) of the input code. The decoder is an important part of the system which selects the cells to be read from and write into. This particular circuit is callexd a decoder matrix, or simply a decoder, and has a characteristic that for each of the possible 2n binary input number which can be taken by the n input cells, the matrix will have a unique one of its 2n output lines selected.

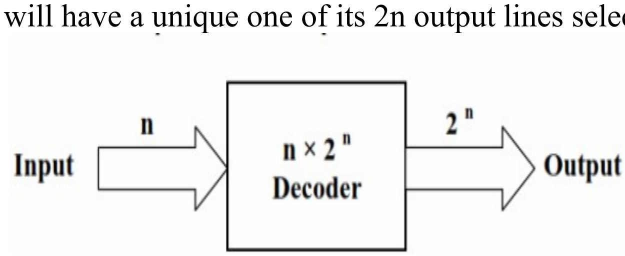

Figure. 7.1: Block Diagram of decoder

The decoder is called n to m where m < 2n for example two-to-four-line decoder. The logic diagram forthe 4-to-2 line decoder is shown in figure 8.1.

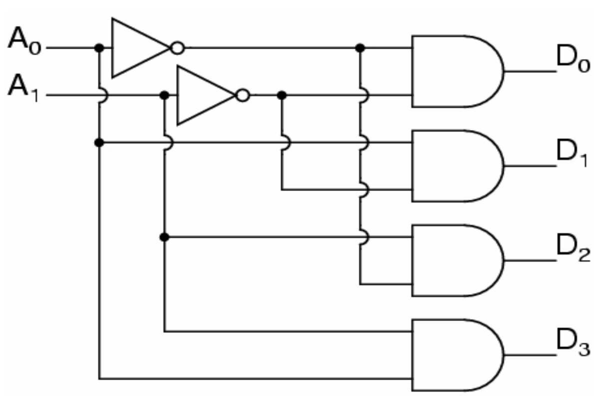

Figure 7.2: 2-to-4 line decoder

Encoder:

An encoder is a combinational logic circuit that generate n output lines from 2n (or less) inputs. It has the reverse function of the decoder. An encoder accepts digit on its inputs, such as a decimal or octal digit, and converts it to a coded output, such as a binary or BCD. Encoder can also be devised to encode various symbol and alphabetic characters. This process of converting from familiar symbols or numbers to a coded format is called encoding.

The logic diagram for the 2-to-4 line encoder is shown in figure 8.2

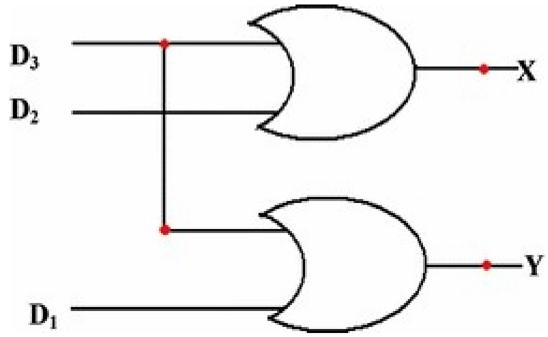

Figure 7.3: 4-to-2 line encoder

Procedure:

- Construct the circuit as shown in logic diagram.
- Connect inputs to switches of logic trainer and output to LED.
- Apply various combinations of inputs and observe the conditions of LEDs.
- Mark results in truth table.

X1 = W3 + W2

X0= W3 + W1

Observations & Calculations:

Table 7.1: Truth table for 4-to-2 line encoder

Inputs				**Outputs**	
W_3	W_2	W_1	W_0	X_1	X_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Truth table for 2-to-4-line decoder

Inputs		**Outputs**			
I_0	I_1	O_1	O_2	O_3	O_4
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Part (b)

Theory:

A priority encoder is a combinational circuit that performs priority. The priority encoder can be implemented by following Boolean function.

$A_0 = D_3 + D_1 D_2'$

$A_1 = D_2 + D_3$

$V = D_0 + D_1 + D_2 + D_3$

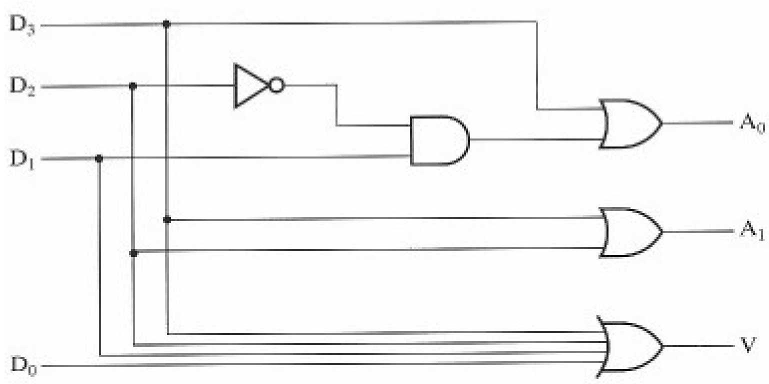

Figure. 7.4: Logic Diagram of encoder

Procedure:

- Construct the circuit as shown in logic diagram.
- Connect inputs to switches of logic trainer and output to LED.
- Apply various combinations of inputs and observe the conditions of LEDs.
- Mark results in truth table.

Observations & Calculations:

$A_0 = D_3 + D_1 D_2'$

$A_1 = D_2 + D_3$

$V = D_0 + D_1 + D_2 + D_3$

Table 7.2: Truth Table of Encoder

D_0	D_1	D_2	D_3	A_1	A_0	$V=D_0+D_1+D_2+D_3$
0	0	0	1	1	1	1
0	0	1	0	1	0	1
0	1	0	0	0	1	1
1	0	0	0	0	0	1
0	0	0	0	0	0	0

Conclusion:

Questions:

1. Design 3 × 8 decoder and verify it through truth table.

3 to 8 Decoder:

- A 3 to 8 decoder has three inputs (A,B,C) and eight outputs (DO to D7).
- Based on the 3 inputs one of the eight outputs is selected.
- The truth table for 3 to 8 decoder is shown in table (1).
- From the truth table, it is seen that only one of eight outputs (DO to D7) is selected based on three select inputs.

From the truth table, the logic expressions for outputs can be written as follows:

A	B	C	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

$$D_0 = \bar{A}\bar{B}\bar{C}, \quad D_1 = \bar{A}\bar{B}C, \quad D_2 = \bar{A}B\bar{C},$$
$$D_3 = \bar{A}BC, \quad D_4 = A\bar{B}\bar{C}, \quad D_5 = A\bar{B}C,$$
$$D_6 = AB\bar{C}, \quad D_7 = ABC$$

- Using the above expressions, the circuit of a 3 to 8 decoder can be implemented using three NOT gates and eight 3-input AND gates as shown in fig (1).
- The three inputs A, B and C are decoded into eight outputs, each output representing one of the midterms of the 3-input variables.
- The three inverters provide the complement of the inputs and each one of the wight AND gates generates one of the midterms.
- This decoder can be used for decoding any 3-bit code to provide eight outputs, corresponding to eight different combinations of the input code.
- This is also called a 1 of 8 decoder, since only one of eight output lines is HIGH for a particular input combination.

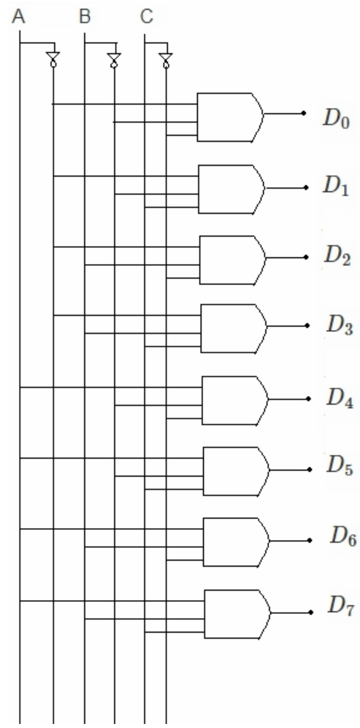

Fig (1): Logic diagram of 3 to 8 decoder.